

END-SEMESTER EXAMINATION, 2024-25
B. Tech. (SEMESTER : 04)
CSE 249 : DATA BASE MANAGEMENT SYSTEM

Time: 3 Hrs.

Max. Marks: 100

- Note:** 1. All questions are compulsory.
 2. Assume missing data suitably, if any.

CO1: Explain the basics concepts of Data Base.

CO2: Demonstrate the knowledge of Databases To E-R Modelling.

CO3: Ability to design entity relationship and convert entity relationship diagrams into RDBMS and formulate SQL Queries on the Respective Data.

CO4: Learn the basic concept of normalization & apply them to reduce redundancy from the Database.

CO5: To appraise the basic issues of transaction processing, Serializability & Concurrency Control

CO6: Design & develop database for real life problems

COs Marks BTL

SECTION-A

All Questions are Compulsory:**(10×4=40 Marks)**

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|---|-----|---|----|
| 1. What is the role of Database Administrator? | CO1 | 4 | K1 |
| 2. Explain three levels of abstraction or schemas architecture of DBMS. | CO1 | 4 | K2 |
| 3. What are aggregate functions? Explain each with an example | CO2 | 4 | K1 |
| 4. Compare WHERE and Having Clause in SQL with example. | CO2 | 4 | K2 |
| 5. Classify all the rules for normalization. Find the highest normal form of relation R (A, B, C, D, E) with FD set as {BC→D, AC→BE, B→E}. If R is not in BCNF, decompose it into a set of BCNF relations that preserve the dependencies. | CO3 | 4 | K2 |
| 6. Apply UNION ALL and INTERSECT Operators with the help of an example | CO3 | 4 | K3 |
| 7. Explain ACID properties of a transaction. | CO4 | 4 | K2 |
| 8. In a distributed transaction processing system, how does the Two-phase Commit Protocol ensure atomicity and what are its limitations in the presence of network failures? | CO4 | 4 | K3 |
| 9. Differentiate between conflict serializability and view serializability with suitable example. | CO5 | 4 | K2 |
| 10. Identify, How Share and exclusive locks differ in terms of access permission? Discuss the concept of Lock Compatibility. | CO5 | 4 | K3 |

SECTION-B

All Questions are Compulsory:**(3×6=18 Marks)**

11. (a) Compare 2NF and 3NF with suitable examples. What types of anomalies are eliminated when relation is in 2 NF? CO3 6 K4

--- OR ---

- (b) Analyse the importance of following terms:

- (i) Relational data model concepts
- (ii) Mapping Constraints
- (iii) Relational Algebra Operations (Join and Division)

12. (a) Explain Wait-Die and Wound-Wait schemes for Deadlock Prevention with CO4 6 K4 examples. Which one is more aggressive, Justify your answer.

--- OR ---

- (b) Explain the difference between cascading rollback and non-cascading schedules. How does cascading rollback affect system performance.
13. (a) Provide examples of scenarios where shared locks are appropriate and where CO5 6 K4 exclusive locks are necessary to maintain data integrity.

--- OR ---

- (b) What is the Optimistic Concurrency Control technique? What is the purpose of validation phase in OCC?

SECTION-C

All Questions are Compulsory:

3×10=30 Marks)

14. (a) Evaluate the following relational DATABASE. Give an expression in SQL CO3 10 K4/K5 for each following queries

Underlined records are Primary Key Employee(person_name , street , city)

Works(person_name, Company_name ,salary)

Company(Company_name , city)

Manages(person_name, manager_name)

- (i) Finds the names and city of all employees who works for the ABC bank.
- (ii) Finds the name of all employees who live in the same city and on the same street as do their managers.
- (iii) Find the name, street address and cities of residence of all employees who work for ABC bank and earn more than 7,000 per annum.
- (iv) Find the name of all employee who earn more than every employee of XYZ.
- (v) Find the names and city of all employees whose names start with 's' and ends with 'z'

--- OR ---

- (b) Explain the concept of Entity Integrity Constraints and Referential Integrity Constraints with a suitable example.
15. (a) Categorise all concurrency control protocol, which protocol ensures both CO4 10 K4/K5 conflict serializability and free from deadlock? Justify your answer.

--- OR ---

- (b) Appraise the timestamp ordering protocol for concurrency control. Differentiate between rigorous and strict 2PL Protocol.
16. (a) For each of the following schedules determine whether the schedule is CO5 10 K4/K5 conflict serializable or view serializable. Justify your answer.
- (i).

T1	T2	T3
		Write (C)
	Write (B)	
	Read (B)	
		Read (C)
Write (C)		

	Read (C)	
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- Is it conflict Serializable? Why?
- Is it View- Serializable? Why?

(ii).

T1	T2	T3
	Read (C)	
Write (C)		
	Write (C)	
Read (B)		
		Write (C)

- Is it conflict Serializable? Why?
- Is it View- Serializable? Why?

--- OR ---

- (b) What do you mean by Database recovery? List various recovery methods. What is Checkpoint in DBMS and How does it help in recovery.

SECTION-D

Attempt the following Question:

(1×12=12 Marks)

17. (a) Explain Immediate Modification Approach and Deferred Modification Approach. 12 K5/K6

Consider the following transaction T1 that transfers 500\$ from Account A to Account B. The initial balances are:

Account A= 3000\$

Account B= 2000\$

Transaction T1 consists of the following steps:

Read (A)

A= A-500

Write (A)

Read (B)

B=B+500

Write(B)

COMMIT

By using Immediate Modification Approach and Deferred Modification Approach-

- What will be the state of Account A and Account B before recovery and after recovery.
- How will the system ensure consistency?

--- OR ---

- (b) Justify with assumptions to draw ER Diagram of University Examination System which demonstrates the following Entities with single value Attributes and Multi-valued attribute and all possible cardinalities.

Entities: University, Student, Course , Department

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